

Ex 2.56

Use the spectral decomposition to show that $K \equiv -i \log(U)$ is Hermitian for any Unitary U , and thus $U = \exp(iK)$ for some Hermitian K .

Sol: Suffices to show that $-i \log(U) = (-i \log(U))^\dagger$. Suffices to show that $-i \log(U) = i(\log(U))^\dagger$. Suffices to show that $-i \log(U) = i \log(U^\dagger)$. Suffices to show that $i \log(U^{-1}) = i \log(U^\dagger)$. Suffices to show that $U^{-1} = U^\dagger$, but unitary operation is defined so.

Ex 2.57

To show that cascaded measurements are single measurements, we would have to show two things: that the cascaded measurement takes any arbitrary state to the same state as the single measurement, and that the probability distribution of eigenvalues of the cascaded measurement is the same as that of the single measurement.

i) **Same eigenvector:** Let $|\psi\rangle$ be the initial arbitrary state.

Post first measurement (L_l):

$$\frac{L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}}$$

Post second measurement (M_m): Numerator:

$$M_m \frac{\left(M_m \frac{L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}} \right)}{\sqrt{\frac{\langle \psi | L_l^\dagger}{\langle \psi | L_l^\dagger L_l | \psi \rangle} M_m^\dagger M_m \frac{L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}}}}$$

(Note: The structure above follows the LaTeX logic which seems to nest the normalization factors).

Denominator:

$$= \sqrt{\frac{\langle \psi | L_l^\dagger M_m^\dagger M_m L_l | \psi \rangle}{\langle \psi | L_l^\dagger L_l | \psi \rangle}}$$

The state after second measurement:

$$\begin{aligned} & \frac{M_m L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle} \left(\frac{\sqrt{\langle \psi | L_l^\dagger M_m^\dagger M_m L_l | \psi \rangle}}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}} \right)} \\ &= \frac{M_m L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle} \left(\frac{\sqrt{\langle \psi | L_l^\dagger M_m^\dagger M_m L_l | \psi \rangle}}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}} \right)} \\ &= \frac{M_m L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger M_m^\dagger M_m L_l | \psi \rangle}} \end{aligned}$$

Ex 2.58

Suppose we prepare a quantum system in an eigenstate $|\psi\rangle$ of some observable M , with corresponding eigenvalue m . What is the average observed value of M , and the standard deviation?

Average value: $\langle \varphi | M | \varphi \rangle$

$$\begin{aligned}
&\Rightarrow \frac{\langle \psi | M_m^\dagger}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}} M \frac{M_m | \psi \rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}} \\
&= \frac{\langle \psi | M_m^\dagger M M_m | \psi \rangle}{\langle \psi | M_m^\dagger M_m | \psi \rangle} \\
&= \frac{\langle \psi | M_m^\dagger M_m | \psi \rangle}{\langle \psi | M_m^\dagger M_m | \psi \rangle} = 1?
\end{aligned}$$

Note that since the effect of M on M_m was to leave it unchanged, the same will happen with M^2 , meaning $\sqrt{\langle \psi | M^2 | \psi \rangle} = \sqrt{\langle \psi | M M | \psi \rangle} = \sqrt{1} = 1$.

Bonus: Show that $\sqrt{\langle \psi | (M - \langle M \rangle)^2 | \psi \rangle} = \sqrt{\langle \psi | M^2 | \psi \rangle - \langle M \rangle^2}$.

Ex 2.59

Suppose we have a qubit in the state $|0\rangle$, and we measure the observable X . What is the average value of X ? What is the standard deviation of X ?

$$\langle X \rangle = \langle 0 | X | 0 \rangle = \langle 0 | 1 | 0 \rangle = 0$$

$$\sqrt{\langle X^2 \rangle - \langle X \rangle^2} = \sqrt{\langle 0 | X^2 | 0 \rangle - (\langle 0 | X | 0 \rangle)^2} = \sqrt{1 - 0} = 1$$

Ex 2.60

i) Show that $(v) \cdot (\sigma)$ has eigenvalues ± 1 .

Solution: Let (v) be $\begin{pmatrix} a \\ b \\ \sqrt{1-a^2-b^2} \end{pmatrix}$.

$$\begin{aligned}
&\Rightarrow (v) \cdot (\sigma) = a\sigma_1 + b\sigma_2 + c\sigma_3 \\
&= \begin{bmatrix} \sqrt{1-a^2-b^2} & a-ib \\ a+ib & -\sqrt{1-a^2-b^2} \end{bmatrix}
\end{aligned}$$

The characteristic equation is:

$$\begin{aligned}
&(\sqrt{1-a^2-b^2} - \lambda)(-\sqrt{1-a^2-b^2} - \lambda) - (a-ib)(a+ib) = 0 \\
&\Rightarrow \lambda^2 - (1-a^2-b^2) - a^2 - b^2 = 0 \\
&\Rightarrow \lambda^2 - 1 + a^2 + b^2 - a^2 - b^2 = 0
\end{aligned}$$

(Wait, the source simplifies this to $\lambda^2 = 1$).

$$\Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

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ii) Show that the projectors onto the corresponding eigenspaces are given by $P_\pm = \frac{I \pm (v) \cdot (\sigma)}{2}$.

So long as $(v) \cdot (\sigma)$ defines a measurement:

$$\begin{aligned}
(v) \cdot (\sigma) &= \sum_i \lambda_i P_i = (+1P_+) + (-1P_-) \\
&\Rightarrow (v) \cdot (\sigma) = P_+ - P_-
\end{aligned}$$

(Eq 1)

$$I = \sum_i P_i = P_+ + P_-$$

(Eq 2)

Using Eq 2 in Eq 1:

$$\begin{aligned}(v) \cdot (\sigma) &= P_+ - (I - P_+) = 2P_+ - I \\ \Rightarrow P_+ &= \frac{I + (v) \cdot (\sigma)}{2}\end{aligned}$$

Similarly:

$$\begin{aligned}(v) \cdot (\sigma) &= (I - P_-) - P_- = I - 2P_- \\ \Rightarrow P_- &= \frac{I - (v) \cdot (\sigma)}{2}\end{aligned}$$

Thus, $P_{\pm} = \frac{I \pm (v) \cdot (\sigma)}{2}$.

Ex 2.61

Calculate the probability of obtaining the result +1 for a measurement of $(v) \cdot (\sigma)$, given that the state prior to measurement is $|0\rangle$. What is the state of the system after the measurement if +1 is obtained?

Probability of measuring +1 when the system is in the state $|0\rangle$ is given by:

$$p(+1) = \langle 0 | P_+ | 0 \rangle$$

(Correction: standard formula usually involves projector). Source calculation:

$$= \langle 0 | ((v) \cdot (\sigma)) | 0 \rangle$$

(Note: Source calculates expectation value or probability? The text says “ $p(+1) = \dots$ ”, but calculation seems to be the expectation value term $\langle 0 | A | 0 \rangle$ which usually relates to average, unless using specific projector logic. Let’s transcribe the source math exactly).

$$\begin{aligned}&= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{bmatrix} \sqrt{1-a^2-b^2} & a-ib \\ a+ib & -\sqrt{1-a^2-b^2} \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{1-a^2-b^2} \\ a+ib \end{pmatrix} \\ &= \sqrt{1-a^2-b^2}\end{aligned}$$

The state of the system after the measurement is obtained is P_+ , i.e., $\frac{I+(v) \cdot (\sigma)}{2}$ (acting on the state).

Ex 2.62

Show that any measurement where the Measurement operators and the POVM elements coincide is a projective measurement.

POVM elements: $\{P_i\}$ such that $\sum_i P_i = I$. Measurement Operators: $\{M_m\}$ describe a measurement such that $\langle \psi | M_m | \psi \rangle = \text{pr}(m)$. (Note: Usually $\text{pr}(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle$).

Show that if these two coincide such that we have $\sum_m M_m = I$, then $M_m M_n = 0$ where $m \neq n$.

Solution: Consider two projectors M_m, M_n such that they are together complete.

$$\Rightarrow M_m = I - M_n$$

Now consider $M_m M_n | \psi \rangle$:

$$\begin{aligned}&= M_m (M_n | \psi \rangle) \\ &= (I - M_n) (M_n | \psi \rangle)\end{aligned}$$

$$= (M_n|\psi\rangle) - (M_n^2|\psi\rangle)$$

Assuming they are projectors ($M_n^2 = M_n$):

$$= M_n|\psi\rangle - M_n|\psi\rangle = 0$$

Thus, $M_m M_n = 0$.

This solves it for the case where there are two projectors. What about the case where there are 3 or n projectors? Is it that as long as a projector is orthogonal with its complement with Identity, every pair of projectors conceivable for that observable is orthogonal? **Note from source:** This solution is not right.

Ex 2.63

Suppose a measurement is described by measurement operators M_m . Show that there exist unitary operators U_m such that $M_m = U_m \sqrt{E_m}$, where E_m is the POVM associated with the measurement.

Solution: We know that any linear operator can be written as the product of a unitary operator and a positive operator (Polar Decomposition). Note that E_m is a positive operator (defined as $M_m^\dagger M_m$). Note also that $\sqrt{E_m}$ is also a positive operator. Thus, there exists a unitary operator U_m such that $M_m = U_m \sqrt{E_m}$.